# The biconstrained function $\psi$ of Chudnovsky, Hompe, Scott, Seymour and Spirkl is not symmetric: $\psi(2/7, 5/7) \leq 23/28 < 6/7 = \psi(5/7, 2/7)$

Machine-generated note (see provenance box)

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**Provenance**

The mathematics in this note was produced without human intervention, in three stages. (1) The constructions and proofs were found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id `1902.10878__01`; original writeup in `attacks/1902.10878__01/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) checked the definitions against the arXiv v3 TeX source of the paper, re-derived every step of the rigidity argument, rebuilt the 42-vertex certificate independently from the prose and verified it exactly (together with the $q = 9, 11$ members of the family and the cyclic constructions for $q = 3, \dots, 13$), and machine-checked the parity obstruction by exhaustive enumeration of all matchings of $K_q$ for $q \leq 13$; its verdict was CONFIRMED. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in Appendix A, the $q = 7$ certificate was written out as explicit tables (Table 1), and Section 4 adds context taken from the referee's report (consistency with the theorems of the source paper, the role of $q \geq 7$, scope). No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

**Abstract**

For $x, y \in (0, 1]$, Chudnovsky, Hompe, Scott, Seymour and Spirkl define $\psi(x, y)$ as the largest $z$ such that in every graph with an $(x, y)$-biconstrained tripartition $(A, B, C)$ some vertex of $C$ is joined by two-edge paths to at least $z|A|$ vertices of $A$. They prove that the one-sided analogue $\phi$ is symmetric, $\phi(x, y) = \phi(y, x)$, and ask whether the same holds for $\psi$. We show that it does not: for every odd $q \geq 7$,

$$\psi\Big(\frac{2}{q}, \frac{q-2}{q}\Big) \;\leq\; 1 - \frac{2(q-2)}{q(q+1)} \;<\; 1 - \frac{1}{q} \;=\; \psi\Big(\frac{q-2}{q}, \frac{2}{q}\Big).$$

The value $\psi((q-2)/q, 2/q) = 1 - 1/q$ is determined exactly by a rigidity argument ending in a parity obstruction; the upper bound on $\psi(2/q, (q-2)/q)$ comes from an explicit triple with $|A| = q(q+1)/2$ and $|B| = |C| = q$. For $q = 7$ this is a 42-vertex graph with $\psi(2/7, 5/7) \leq 23/28 < 24/28 = \psi(5/7, 2/7)$.

## 1 Introduction

All graphs are finite and simple. Following [1], a *tripartition* of a graph $G$ is a partition $(A, B, C)$ of $V(G)$ into three nonempty stable sets such that there are no edges between $A$ and $C$. For $x, y \in (0, 1]$ the tripartition is $(x, y)$-*constrained* if every vertex of $A$ has at least $x|B|$ neighbours in $B$ and every vertex of $B$ has at least $y|C|$ neighbours in $C$, and it is $(x, y)$-*biconstrained* if in addition every vertex of $B$ has at least $x|A|$ neighbours in $A$ and every vertex of $C$ has at least $y|B|$ neighbours in $B$. In symbols, writing $N_X(v)$ for the set of neighbours of $v$ in $X$, the four biconstrained conditions read

$$\begin{aligned} &|N_B(a)| \geq x|B| \ \ (a \in A), && |N_A(b)| \geq x|A| \ \ (b \in B), \\ &|N_C(b)| \geq y|C| \ \ (b \in B), && |N_B(c)| \geq y|B| \ \ (c \in C). \end{aligned} \tag{1}$$

For $c \in C$ let $N_A^2(c)$ be the set of vertices of $A$ at distance exactly two from $c$; since the parts are stable and there are no $A$–$C$ edges, this is

$$N_A^2(c) = \mathrm{R}(c) := \{a \in A : \ N_B(a) \cap N_B(c) \neq \emptyset\}.$$

The paper defines $\phi(x, y)$ as the maximum $z$ such that every $(x, y)$-constrained tripartition has a vertex $c \in C$ with $|N_A^2(c)| \geq z|A|$, and $\psi(x, y)$ in the same way for $(x, y)$-biconstrained tripartitions (the maximum exists, [1, Theorem 2.1]); equivalently

$$\psi(x, y) = \inf_{(G,A,B,C)} \max_{c \in C} \frac{|\mathrm{R}(c)|}{|A|}, \tag{2}$$

the infimum over all $(x, y)$-biconstrained tripartitions. Theorem 2.3 of [1] states that $\phi(x, y) = \phi(y, x)$ for all $x, y \in (0, 1]$, proved by a linear-programming argument. Immediately afterwards the authors write: “we have not been able to prove an analogue of [Theorem 2.3] for the biconstrained case, or for the exact case, although we have no counterexample for either one.” The catalog records this as:

**Question 1.1** ([1], remark after Theorem 2.3)**.** Does $\psi(x, y) = \psi(y, x)$ hold for all $x, y \in (0, 1]$?

We answer this negatively.

**Theorem 1.2** (Main theorem)**.** *For every odd integer $q \geq 7$,*

$$\psi\Big(\frac{2}{q}, \frac{q-2}{q}\Big) \ \leq \ 1 - \frac{2(q-2)}{q(q+1)} \ < \ 1 - \frac{1}{q} \ = \ \psi\Big(\frac{q-2}{q}, \frac{2}{q}\Big).$$

**Corollary 1.3.** $\psi(2/7, 5/7) \leq \frac{23}{28} < \frac{24}{28} = \frac{6}{7} = \psi(5/7, 2/7)$. *In particular the answer to Question 1.1 is negative.*

*Proof.* Take $q = 7$ in Theorem 1.2: $1 - \frac{2\cdot5}{7\cdot8} = 1 - \frac{10}{56} = \frac{23}{28}$ and $1 - \frac{1}{7} = \frac{24}{28}$. □

Theorem 1.2 combines two statements, proved in Sections 2 and 3: the exact value $\psi((q-2)/q, 2/q) = 1 - 1/q$ for all odd $q \geq 3$ (Lemma 2.1), and the upper bound $\psi(2/q, (q-2)/q) \leq 1 - 2(q-2)/(q(q+1))$ for odd $q \geq 7$ (Proposition 3.1). The theorem follows because

$$\frac{2(q-2)}{q(q+1)} - \frac{1}{q} = \frac{2q - 4 - (q+1)}{q(q+1)} = \frac{q-5}{q(q+1)} > 0 \qquad (q > 5). \tag{3}$$

*Idea of proof.* Both values are governed by the same rigidity: if $a \in A$ is *not* reached by $c \in C$, then $N_B(a)$ and $N_B(c)$ are disjoint, and when $x + y = 1$ the degree conditions force $N_B(a) = B \setminus N_B(c)$ with both degrees tight. For $(x, y) = ((q-2)/q, 2/q)$ this rigidity propagates: if every $c$ missed more than $|A|/q$ vertices, the $B$-neighbourhoods $S_i$ of $C$-vertices would each carry exactly a $2/q$ fraction of $C$, and the $B$-side degree bound would force these $S_i$ to be pairwise disjoint, so that $m$ of them cover $C$ exactly, giving $2m = q$, which is impossible for odd $q$. A cyclic construction on $3q$ vertices shows $1 - 1/q$ is attained. For $(x, y) = (2/q, (q-2)/q)$ the roles flip: now $A$-vertices have tiny neighbourhoods $S_i$ of size two, and there is no parity obstruction to choosing $m = (q+3)/2$ two-element sets $S_i$ in a $q$-set $B$ that “cover” $B$ with the right multiplicities (disjoint pairs plus a triangle). Giving each $S_i$ a block of $q-2$ $A$-vertices, adding three $A$-vertices complete to $B$ to lift the $B$-side degrees, and letting each $C$-vertex miss exactly one $S_i$ produces a triple in which every $C$-vertex misses exactly $q-2$ of the $q(q+1)/2$ vertices of $A$. *[full proof]*

## 2 The exact value of $\psi((q-2)/q,\, 2/q)$

**Lemma 2.1.** *For every odd integer $q \geq 3$, $\psi\Big(\dfrac{q-2}{q}, \dfrac{2}{q}\Big) = 1 - \dfrac{1}{q}$.*

*Idea of proof. Lower bound.* Suppose every $c \in C$ reaches fewer than $\frac{q-1}{q}|A|$ vertices. For an unreached pair $(a, c)$, disjointness of $N_B(a), N_B(c)$ and $|N_B(a)| + |N_B(c)| \geq |B|$ force $N_B(a) = B \setminus N_B(c)$ and $|N_B(c)| = \frac{2}{q}|B|$. Counting $B$–$C$ edges then forces $|N_C(b)| = \frac{2}{q}|C|$ for every $b$. Group $C$ by its $B$-neighbourhoods $S_1, \dots, S_m$, with fractions $\beta_i$ of $C$, and let $A_i$ be the $A$-vertices with neighbourhood $B \setminus S_i$, with fractions $\alpha_i > 1/q$ of $A$. For $b \in B$, the indices $i$ with $b \in S_i$ have $\sum \beta_i = 2/q$ and, because $b$ has at most $\frac{2}{q}|A|$ non-neighbours in $A$, $\sum \alpha_i \leq 2/q$; hence each $b$ lies in exactly one $S_i$, each $\beta_i = 2/q$, and $1 = \sum \beta_i = 2m/q$, contradicting the parity of $q$. *Upper bound.* On $A = B = C = \mathbb{Z}_q$ let $N_B(c_i) = \{b_i, b_{i+1}\}$ and $N_B(a_i) = B \setminus \{b_i, b_{i+1}\}$; each $c_i$ misses exactly $a_i$. *[full proof]*

## 3 The construction for $\psi(2/q,\, (q-2)/q)$

Let $q \geq 7$ be odd and put

$$h = \frac{q-3}{2}\ (\geq 2), \qquad m = h + 3 = \frac{q+3}{2}.$$

Let $B$ consist of $h$ pairwise disjoint pairs $P_1, \dots, P_h$ together with three further vertices $r_{12}, r_{13}, r_{23}$, so $|B| = 2h + 3 = q$. Define $m$ distinct two-element subsets of $B$:

$$S_i = P_i \quad (1 \leq i \leq h), \qquad S_{h+1} = \{r_{12}, r_{13}\}, \quad S_{h+2} = \{r_{12}, r_{23}\}, \quad S_{h+3} = \{r_{13}, r_{23}\}.$$

- $A$: for each $i \in \{1, \dots, m\}$ a set $A_i$ of $q-2$ vertices, each with $N_B(a) = S_i$; and a set $U$ of three vertices, each with $N_B(u) = B$. Thus $|A| = m(q-2) + 3 = q(q+1)/2$.
- $C$: for each $1 \leq i \leq h$ two vertices *of type* $i$, and for each $i \in \{h+1, h+2, h+3\}$ one vertex of type $i$; a vertex $c$ of type $i$ has $N_B(c) = B \setminus S_i$. Thus $|C| = 2h + 3 = q$.

The graph $G_q$ has vertex set $A \cup B \cup C$ and exactly the $A$–$B$ and $B$–$C$ edges just described; so $(A, B, C)$ is a tripartition (three nonempty stable sets, no $A$–$C$ edges).

**Proposition 3.1.** *For every odd $q \geq 7$, $(A, B, C)$ is a $(2/q, (q-2)/q)$-biconstrained tripartition of $G_q$ with $|A| = q(q+1)/2$, $|B| = |C| = q$, and every $c \in C$ satisfies $|A \setminus \mathrm{R}(c)| = q - 2$. Consequently*

$$\psi\Big(\frac{2}{q}, \frac{q-2}{q}\Big) \leq 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}.$$

*Idea of proof.* Degrees: $A_i$-vertices have $2 = \frac{2}{q}|B|$ neighbours in $B$; a vertex of a pair $P_i$ is adjacent to $A_i \cup U$, i.e. to $q+1 = \frac{2}{q}|A|$ vertices; each $r_{jk}$ lies in two of the last three $S_i$ and has $2(q-2)+3 \geq q+1$; every $c$ has $q - 2 = \frac{q-2}{q}|B|$ neighbours in $B$; every $b$ is omitted by exactly two $C$-vertices, hence has $q - 2 = \frac{q-2}{q}|C|$ neighbours in $C$. Reach: a $c$ of type $i$ misses $A_i$ (whose neighbourhood is $S_i$) and reaches every $A_j$, $j \neq i$ (the distinct two-sets satisfy $S_j \not\subseteq S_i$) and $U$. *[full proof]*

**The certificate for** $q = 7$. Here $h = 2$, $m = 5$, $|A| = 28$, $|B| = |C| = 7$. Write $B = \{1, \dots, 7\}$ with $P_1 = \{1, 2\}$, $P_2 = \{3, 4\}$, $r_{12} = 5$, $r_{13} = 6$, $r_{23} = 7$, so

$$S_1 = \{1, 2\}, \quad S_2 = \{3, 4\}, \quad S_3 = \{5, 6\}, \quad S_4 = \{5, 7\}, \quad S_5 = \{6, 7\}.$$

Table 1 lists every vertex of $A$ and $C$ with its neighbourhood in $B$; there are no other edges. Every $C$-vertex misses exactly the five $A$-vertices of its own type and reaches the other 23, so $\psi(2/7, 5/7) \leq 23/28$.

| part | vertices | $N_B(\cdot)$ | count |
|---|---|---|---|
| $A$ | $a_{1,1}, \dots, a_{1,5}$ | $\{1,2\}$ | 5 |
| | $a_{2,1}, \dots, a_{2,5}$ | $\{3,4\}$ | 5 |
| | $a_{3,1}, \dots, a_{3,5}$ | $\{5,6\}$ | 5 |
| | $a_{4,1}, \dots, a_{4,5}$ | $\{5,7\}$ | 5 |
| | $a_{5,1}, \dots, a_{5,5}$ | $\{6,7\}$ | 5 |
| | $u_1, u_2, u_3$ | $\{1,2,3,4,5,6,7\}$ | 3 |
| $C$ | $c_1, c_1'$ | $\{3,4,5,6,7\}$ | 2 |
| | $c_2, c_2'$ | $\{1,2,5,6,7\}$ | 2 |
| | $c_3$ | $\{1,2,3,4,7\}$ | 1 |
| | $c_4$ | $\{1,2,3,4,6\}$ | 1 |
| | $c_5$ | $\{1,2,3,4,5\}$ | 1 |

Table 1: The 42-vertex $(2/7, 5/7)$-biconstrained triple, $B = \{1, \dots, 7\}$. Degree check: each $a_{i,j}$ has $2 = \frac{2}{7} \cdot 7$ neighbours in $B$; each $b \in \{1,2,3,4\}$ has $5 + 3 = 8 = \frac{2}{7} \cdot 28$ neighbours in $A$ and each $b \in \{5,6,7\}$ has $10 + 3 = 13$; each $c$ has $5 = \frac{5}{7} \cdot 7$ neighbours in $B$; each $b$ is missing from exactly two rows of the $C$ block, so has $5 = \frac{5}{7} \cdot 7$ neighbours in $C$. Reach check: $c$ of type $i$ has a common $B$-neighbour with every $A$-vertex except the five with neighbourhood $S_i$.

## 4 Remarks

*Remark* 4.1 (Why $q \geq 7$). By (3) the two bounds of Theorem 1.2 differ by $(q-5)/(q(q+1))$, which vanishes at $q = 5$ and is negative at $q = 3$. At $q = 3$ both sides equal $2/3$, in agreement with the values $\psi(1/3, 2/3) = \psi(2/3, 1/3) = 2/3$ that follow from the source paper's Theorems 2.2 and the "$2+3$" bounds (as the referee notes); the construction of Section 3 needs $h \geq 2$ anyway, i.e. $q \geq 7$.

*Remark* 4.2 (Consistency with the source paper). The referee (Appendix B) checked the two values against every result of [1]. The trivial bounds $\max(x, y) \leq \psi(x, y) \leq (\lceil kx \rceil + \lceil ky \rceil - 1)/k$ give, with $k = 7$, $\psi \in [5/7, 6/7]$ for both $(5/7, 2/7)$ and $(2/7, 5/7)$; the value $6/7$ attains the upper bound, and $23/28$ lies strictly inside. The paper's Theorem giving $\psi(x, y) \geq 3/4$ for $x \geq 1/4$, $y > 2/3$ yields $\psi(2/7, 5/7) \geq 3/4 = 21/28$, so $\psi(2/7, 5/7) \in [21/28, 23/28]$; its exact value is not determined here. The paper's partial symmetry result (Theorem 10.2: $\psi(x, y) = x$ iff $\psi(y, x) = x$) is not contradicted, since neither value equals $\max(x, y)$.

*Remark* 4.3 (Scope). The symmetry $\phi(x, y) = \phi(y, x)$ (Theorem 2.3 of [1]) is unaffected. The construction uses vertices of $A$ complete to $B$, so it says nothing about the paper's *exact* variant (every degree condition holding with equality), whose symmetry question remains open.

*Remark* 4.4 (Computational checks). The referee rebuilt both constructions from the prose and verified, with exact rational arithmetic, all degree conditions and all second neighbourhoods for $q = 7, 9, 11$ (every $c$ reaches $23/28$, $38/45$, $57/66$ of $A$) and for the cyclic construction for $q = 3, \dots, 13$; it also verified the finite core of the parity step by enumerating all matchings of $K_q$, $q \leq 13$. Scripts: `verification/scripts/1902.10878__01/`.

*Remark* 4.5 (Novelty). The referee's literature search (September 2026) found no resolution of Question 1.1; the published version of [1] still lists it as open, and the exact value $\psi((q-2)/q, 2/q) = 1 - 1/q$ does not appear there. This has not been checked by a human.

## A Full proofs

**Lemma 2.1.** *For every odd integer $q \geq 3$, $\psi\Big(\dfrac{q-2}{q}, \dfrac{2}{q}\Big) = 1 - \dfrac{1}{q}$.*

*Proof.* Fix odd $q \geq 3$ and write $x = (q-2)/q$, $y = 2/q$, so $x + y = 1$.

*Lower bound.* Let $(A,B,C)$ be an $(x,y)$-biconstrained tripartition of a graph $G$, and suppose for a contradiction that

$$|\mathrm{R}(c)| < \frac{q-1}{q}|A| \qquad \text{for every } c \in C. \tag{4}$$

Then every $c \in C$ has more than $|A|/q > 0$ vertices of $A$ outside $\mathrm{R}(c)$. Let $a \notin \mathrm{R}(c)$. Then $N_B(a) \cap N_B(c) = \emptyset$, so by (1)

$$|B| \;\geq\; |N_B(a)| + |N_B(c)| \;\geq\; \frac{q-2}{q}|B| + \frac{2}{q}|B| \;=\; |B|.$$

Equality holds throughout, hence

$$|N_B(a)| = \frac{q-2}{q}|B|, \qquad |N_B(c)| = \frac{2}{q}|B|, \qquad N_B(a) = B \setminus N_B(c) \qquad \text{whenever } a \notin \mathrm{R}(c). \tag{5}$$

Since every $c$ has some unreached $a$, (5) gives $|N_B(c)| = \frac{2}{q}|B|$ for every $c \in C$, so $e(B,C) = \sum_{c\in C} |N_B(c)| = \frac{2}{q}|B||C|$. On the other hand $e(B,C) = \sum_{b\in B} |N_C(b)|$ with every term at least $\frac{2}{q}|C|$ by (1); equality of the totals forces

$$|N_C(b)| = \frac{2}{q}|C| \qquad \text{for every } b \in B. \tag{6}$$

Let $S_1, \dots, S_m$ be the distinct sets occurring as $N_B(c)$, $c \in C$, and put

$$C_i = \{c \in C : N_B(c) = S_i\}, \quad \beta_i = \frac{|C_i|}{|C|}, \qquad A_i = \{a \in A : N_B(a) = B \setminus S_i\}, \quad \alpha_i = \frac{|A_i|}{|A|}.$$

Then $\beta_i > 0$, $\sum_i \beta_i = 1$, and the sets $A_i$ are pairwise disjoint because the $S_i$ are distinct. For $c \in C_i$ we claim $A \setminus \mathrm{R}(c) = A_i$: if $a \notin \mathrm{R}(c)$ then $N_B(a) = B \setminus S_i$ by (5), so $a \in A_i$; conversely if $a \in A_i$ then $N_B(a) \cap S_i = \emptyset$, so $a \notin \mathrm{R}(c)$. Hence (4) gives

$$\alpha_i > \frac{1}{q} \qquad (1 \leq i \leq m). \tag{7}$$

For $b \in B$ let $I(b) = \{i : b \in S_i\}$. Since $N_C(b) = \bigcup_{i\in I(b)} C_i$ (a disjoint union), (6) gives

$$\sum_{i\in I(b)} \beta_i = \frac{2}{q}, \tag{8}$$

in particular $I(b) \neq \emptyset$. If $i \in I(b)$, every $a \in A_i$ has $N_B(a) = B \setminus S_i \not\ni b$, so $A_i$ is entirely non-adjacent to $b$. By (1), $b$ has at least $\frac{q-2}{q}|A|$ neighbours in $A$, hence at most $\frac{2}{q}|A|$ non-neighbours, and since the $A_i$ are disjoint,

$$\sum_{i\in I(b)} \alpha_i \leq \frac{2}{q}. \tag{9}$$

By (7), two terms of this sum would already exceed $2/q$; so $|I(b)| = 1$ for every $b \in B$, and if $I(b) = \{i\}$ then (8) gives $\beta_i = 2/q$. Every index $i$ occurs in this way: $S_i = N_B(c)$ for some $c$ has size $\frac{2}{q}|B| > 0$, and any $b \in S_i$ has $I(b) = \{i\}$. Therefore $\beta_i = 2/q$ for all $i$ and

$$1 = \sum_{i=1}^{m} \beta_i = \frac{2m}{q},$$

i.e. $q = 2m$, contradicting that $q$ is odd. Hence some $c \in C$ has $|\mathrm{R}(c)| \geq \frac{q-1}{q}|A|$, and by (2) $\psi(x,y) \geq 1 - \frac{1}{q}$.

*Upper bound.* Let $A = \{a_i : i \in \mathbb{Z}_q\}$, $B = \{b_i : i \in \mathbb{Z}_q\}$, $C = \{c_i : i \in \mathbb{Z}_q\}$, put $S_i = \{b_i, b_{i+1}\}$ (indices modulo $q$; $b_i \neq b_{i+1}$ as $q \geq 3$), and let the graph have exactly the edges given by

$$N_B(a_i) = B \setminus S_i, \qquad N_B(c_i) = S_i \qquad (i \in \mathbb{Z}_q).$$

This is a tripartition. Each $a_i$ has $q-2 = x|B|$ neighbours in $B$; $b_j$ lies in $S_j$ and $S_{j-1}$ only, so it is non-adjacent to exactly $a_j, a_{j-1}$ and has $q-2 = x|A|$ neighbours in $A$; each $c_i$ has $2 = y|B|$ neighbours in $B$; and $b_j$ has exactly the two neighbours $c_j, c_{j-1}$ in $C$, i.e. $2 = y|C|$. So the triple is $(x,y)$-biconstrained. The vertex $c_i$ does not reach $a_i$, since $N_B(a_i) \cap S_i = \emptyset$. For $j \neq i$ the two-element sets $S_i \neq S_j$ satisfy $S_i \setminus S_j \neq \emptyset$, so $N_B(c_i) \cap N_B(a_j) = S_i \setminus S_j \neq \emptyset$ and $c_i$ reaches $a_j$. Hence every $c_i$ reaches exactly $q-1$ vertices of $A$, and $\psi(x,y) \leq 1 - \frac{1}{q}$. □

**Proposition 3.1.** *For every odd $q \geq 7$, $(A,B,C)$ is a $(2/q, (q-2)/q)$-biconstrained tripartition of $G_q$ with $|A| = q(q+1)/2$, $|B| = |C| = q$, and every $c \in C$ satisfies $|A \setminus \mathrm{R}(c)| = q-2$. Consequently*

$$\psi\Big(\frac{2}{q}, \frac{q-2}{q}\Big) \leq 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}.$$

*Proof.* Throughout, $x = 2/q$ and $y = (q-2)/q$. First, $|B| = 2h+3 = q$ and $|C| = 2h+3 = q$ by construction, and

$$|A| = m(q-2) + 3 = \frac{(q+3)(q-2)}{2} + 3 = \frac{q^2+q-6}{2} + 3 = \frac{q(q+1)}{2}.$$

The sets $S_1, \dots, S_m$ are pairwise distinct two-element subsets of $B$: the pairs $P_i$ are disjoint from each other and from $\{r_{12}, r_{13}, r_{23}\}$, and the three sets $S_{h+1}, S_{h+2}, S_{h+3}$ are the three two-subsets of $\{r_{12}, r_{13}, r_{23}\}$.

*A–B conditions.* A vertex $a \in A_i$ has $|N_B(a)| = |S_i| = 2 = \frac{2}{q}|B| = x|B|$; a vertex $u \in U$ has $|N_B(u)| = q \geq x|B|$. A vertex $b \in P_i$ belongs to $S_i$ and to no other $S_j$, so $N_A(b) = A_i \cup U$ and $|N_A(b)| = (q-2)+3 = q+1 = \frac{2}{q} \cdot \frac{q(q+1)}{2} = x|A|$. A vertex $r_{jk}$ belongs to exactly two of $S_{h+1}, S_{h+2}, S_{h+3}$ (namely those containing $r_{jk}$) and to no $P_i$, so $|N_A(r_{jk})| = 2(q-2)+3 = 2q-1 \geq q+1 = x|A|$.

*B–C conditions.* A vertex $c$ of type $i$ has $|N_B(c)| = |B \setminus S_i| = q-2 = \frac{q-2}{q}|B| = y|B|$. A vertex $b \in B$ is non-adjacent to $c$ exactly when $b \in S_{\text{type}(c)}$. If $b \in P_i$, this happens exactly for the two vertices of type $i$; if $b = r_{jk}$, it happens for the two types among $h+1, h+2, h+3$ whose set contains $r_{jk}$, each of which occurs once. So every $b$ is non-adjacent to exactly two vertices of $C$ and $|N_C(b)| = q-2 = y|C|$. Thus $(A,B,C)$ is $(x,y)$-biconstrained.

*Second neighbourhoods.* Let $c$ have type $i$, so $N_B(c) = B \setminus S_i$. A vertex $a \in A_i$ has $N_B(a) = S_i$, disjoint from $N_B(c)$, so $a \notin \mathrm{R}(c)$. For $j \neq i$, $S_j \neq S_i$ are two-element sets, so $S_j \setminus S_i \neq \emptyset$, i.e. every $a \in A_j$ has a common neighbour with $c$ in $B \setminus S_i$; and every $u \in U$ is adjacent to all of $B \setminus S_i \neq \emptyset$. Hence $A \setminus \mathrm{R}(c) = A_i$ and $|A \setminus \mathrm{R}(c)| = q-2$, for every $c \in C$. Every $c$ therefore reaches the fraction

$$\frac{|A| - (q-2)}{|A|} = 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}$$

of $A$, and (2) gives the stated upper bound on $\psi(x,y)$. □

**Theorem 1.2** (Main theorem)**.** *For every odd integer $q \geq 7$,*

$$\psi\Big(\frac{2}{q}, \frac{q-2}{q}\Big) \;\leq\; 1 - \frac{2(q-2)}{q(q+1)} \;<\; 1 - \frac{1}{q} \;=\; \psi\Big(\frac{q-2}{q}, \frac{2}{q}\Big).$$

*Proof.* Let $q \geq 7$ be odd. Proposition 3.1 gives $\psi(2/q, (q-2)/q) \leq 1 - \frac{2(q-2)}{q(q+1)}$ and Lemma 2.1 gives $\psi((q-2)/q, 2/q) = 1 - \frac{1}{q}$. By (3), $\frac{2(q-2)}{q(q+1)} > \frac{1}{q}$ for $q > 5$, so $1 - \frac{2(q-2)}{q(q+1)} < 1 - \frac{1}{q}$. □

# B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (`claude-fable-5`, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/1902.10878__01/`. Treat it as machine output, not as an authority.

Claim under review: the writeup at `attacks/1902.10878__01/output.md` claims to **disprove** the open question "does psi(x,y) = psi(y,x) for all x,y in (0,1]?" from arXiv:1902.10878 (Chudnovsky, Hompe, Scott, Seymour, Spirkl, *Concatenating bipartite graphs*), via

- a Lemma: psi((q-2)/q, 2/q) = 1 - 1/q for every odd q >= 3, and
- a construction: psi(2/q, (q-2)/q) <= 1 - 2(q-2)/(q(q+1)) for odd q >= 7,

which for q = 7 gives psi(2/7,5/7) <= 23/28 < 24/28 = 6/7 = psi(5/7,2/7).

## B.1 Interpretation

I downloaded the arXiv v3 TeX source (Dec 2020, the version into which the withdrawn companion arXiv:1908.07453 was merged) and checked the definitions verbatim.

- Paper definition (source lines 101–110, 179–196): a *tripartition* (A,B,C) is a partition of V(G) into nonempty **stable** sets; (x,y)-constrained means every a in A has >= x|B| neighbours in B, every b in B has >= y|C| neighbours in C, and **no edges between A and C**; (x,y)-*biconstrained* adds: every b in B has >= x|A| neighbours in A and every c in C has >= y|B| neighbours in B. psi(x,y) is the maximum z such that every (x,y)-biconstrained G has some v in C with |N^2_A(v)| >= z|A|, where N^2 is distance **exactly** two.
- The writeup's four degree conditions match the paper's exactly (same sides, same fractions, no silent strengthening or weakening). Its R(c) = {a : N_B(a) and N_B(c) intersect} equals N^2_A(c): in a tripartition N(a), N(c) are subsets of B and there are no A–C edges, so distance exactly 2 from c to a is equivalent to a common B-neighbour. The writeup's constructions have all three parts stable, no A–C edges, and V = A + B + C, so they are legitimate tripartitions.
- The open question is stated in the paper as a prose remark right after Theorem 2.3 (`permute`, phi(x,y)=phi(y,x)), source line 437: "we have not been able to prove an analogue of [2.3] for the biconstrained case, or for the exact case, although we have no counterexample for either one." The catalog question ("Does psi(x,y)=psi(y,x) hold for all x,y in (0,1]?") is exactly what the writeup answers, in the negative, with a concrete counterexample pair. No interpretation gaming: a reasonable author of the original paper would consider the psi-symmetry question resolved (negatively) by this argument. The writeup correctly does not claim anything about the exact-case (xi) symmetry, which remains open.

Interpretation: **the intended one**.

## B.2 Step-by-step findings

<table>
<tr><th>step</th><th>label</th><th>note</th></tr>
<tr><td>1.1 Definitions of biconstrained and psi</td><td>VALID</td><td>Matches arXiv v3 verbatim (lines 101–110, 179–196 of source). inf/max formulation equivalent to the paper's "maximum z" (paper proves the max exists, Thm <code>suptomax</code>).</td></tr>
<tr><td>2.1 Lemma lower bound: disjointness forces equality chain (1)</td><td>VALID</td><td>If a is not in R(c), N_B(a), N_B(c) disjoint; degree sums give |B| >= ((q-2)/q + 2/q)|B| = |B|, so all equalities and N_B(a) = B  N_B(c). Under the contradiction hypothesis every c has > |A|/q >= 1 unreachable a, so every c has exactly (2/q)|B| B-neighbours. (Divisibility is not an issue: equality forced means (2/q)|B| is an integer; if it could not be, no c has an unreachable vertex, contradicting the hypothesis directly.)</td></tr>
<tr><td>2.2 Edge count gives (2): every b has exactly (2/q)|C| C-neighbours</td><td>VALID</td><td>e(B,C) = (2/q)|B||C| from the C side; each of the |B| terms on the B side is >= (2/q)|C|; equality in the total forces equality termwise.</td></tr>
<tr><td>2.3 Classes C_i, A_i and (3): alpha_i > 1/q</td><td>VALID</td><td>For c in C_i the unreachable set is exactly A_i = {a : N_B(a) = B  S_i} (both inclusions immediate from (1)); the A_i are pairwise disjoint (distinct S_i); hypothesis |R(c)| < (q-1)|A|/q gives |A_i| > |A|/q.</td></tr>
<tr><td>2.4 (4): sum of beta_i over i in I(b) equals 2/q</td><td>VALID</td><td>{c : b in N_B(c)} = N_C(b), which has size (2/q)|C| by (2). Hence I(b) nonempty.</td></tr>
<tr><td>2.5 (5): sum of alpha_i over i in I(b) at most 2/q</td><td>VALID</td><td>b in S_i means A_i is entirely nonadjacent to b (N_B(a) = B  S_i); b has >= (q-2)|A|/q A-neighbours (biconstrained, x = (q-2)/q), so at most 2|A|/q nonneighbours; the A_i are disjoint.</td></tr>
</table>

| step | label | note |
|---|---|---|
| 2.6 Parity contradiction: $\lvert I(b)\rvert = 1$, $\beta_i = 2/q$, $2m = q$ | VALID | Two alpha-terms each $> 1/q$ would exceed $2/q$, so $\lvert I(b)\rvert = 1$; then (4) gives $\beta_i = 2/q$; every $i$ occurs ($S_i$ nonempty, pick $b$ in $S_i$); summing, $1 = 2m/q$, impossible for odd $q$. Finite core machine-verified (script `verify_lemma_core.py`): no matching of $K_q$ covers all $q$ vertices for $q = 3..13$ (exhaustive enumeration, e.g. 232 matchings of $K_7$, 568504 of $K_{13}$; sanity check $q = 6$ does have one). Hence $\psi((q-2)/q, 2/q) >= 1 - 1/q$. |
| 2.7 Lemma upper bound: cyclic construction | VALID | Verified by computer for $q = 3,5,7,9,11,13$: the triple is exactly $((q-2)/q, 2/q)$-biconstrained and every $c$ in $C$ has $\lvert N^2_A(c)\rvert = q-1$, so $\psi((q-2)/q, 2/q) <= 1 - 1/q$. Together with 2.6: equality. |
| 3.1 Set B: $h$ pairs plus r12, r13, r23; $m = (q+3)/2$ distinct 2-subsets $S_i$ | VALID | $\lvert B\rvert = 2h + 3 = q$; the $m$ 2-subsets are pairwise distinct (disjoint pairs; the three r-sets pairwise distinct and supported off the pairs). Verified in code. |
| 3.2 Set A and the A–B side, $\lvert A\rvert = q(q+1)/2$ | VALID | $\lvert A\rvert = m(q-2) + 3 = q(q+1)/2$ (exact-fraction check passed). $A_i$-vertices have exactly $2 = (2/q)\lvert B\rvert$ B-neighbours; pair-vertices have exactly $q+1 = (2/q)\lvert A\rvert$ A-neighbours (tight); r-vertices have $2q-1 >= q+1$. Verified by computer for $q = 7, 9, 11$. |
| 3.3 Set C and the B–C side | VALID | $\lvert C\rvert = 2h + 3 = q$; every $c$ has $q-2 = ((q-2)/q)\lvert B\rvert$ B-neighbours; every $b$ is omitted by exactly two C-vertices (pair-vertices by the two type-$i$ copies; r-vertices by the two singleton types containing them), so has $q-2 = ((q-2)/q)\lvert C\rvert$ C-neighbours. Verified by computer. The triple is $(2/q,(q-2)/q)$-biconstrained. |

| step | label | note |
|---|---|---|
| 3.4 Second neighbourhoods and final inequality | VALID | c of type i misses exactly the q-2 vertices of A_i (S_j  S_i nonempty for j != i since the S's are distinct 2-sets; the 3 universal vertices are always reached). Computer: for q = 7/9/11 every c reaches exactly 23/38/57 of \|A\| = 28/45/66. And 2(q-2)/(q(q+1)) - 1/q = (q-5)/(q(q+1)) > 0 iff q > 5 (exact-fraction check). |
| 4 q = 7 instance | VALID | Independently rebuilt: \|A\| = 28, \|B\| = \|C\| = 7, biconstrained for (2/7, 5/7), every C-vertex reaches exactly 23 of 28 A-vertices; 23/28 < 24/28 = 6/7. |
| Verdict block ("disproved", one_line, caveats) | VALID | Accurate: symmetry fails at (x,y) = (2/7, 5/7); the writeup correctly notes psi(2/7,5/7) is not determined exactly and that the exact-degree (xi) question is untouched. |

No GAP or ERROR found. The only step that is not a finite object — the Lemma's lower bound (2.1–2.6), a universal statement over all biconstrained graphs — was re-derived line by line above and its combinatorial core (the forced matching structure and the parity obstruction) machine-checked.

## B.3 Reference check

The writeup cites no external theorems; everything is self-contained except the definitions, which I verified against the downloaded arXiv v3 TeX source of 1902.10878 (identical to the published EJC version, v29i2p47, 2022, per the journal listing). Additional due diligence — consistency of the two claimed values with everything the source paper proves:

- `trivialbounds` (paper Thm 2.2): max(x,y) <= psi(x,y) <= (ceil(kx)+ceil(ky)-1)/k. For (5/7, 2/7), k = 7: psi in [5/7, 6/7]; the Lemma's value 6/7 attains the upper (Cayley) bound — consistent. For (2/7, 5/7): psi in [5/7, 6/7], and 23/28 lies strictly inside — consistent.
- `3/4ub2` (x > 2/3, x+2y > 1, 4(1-x)(1-y) <= 1): gives psi(5/7,2/7) >= 3/4 (40/49 <= 1); consistent with (and weaker than) the Lemma's 6/7.
- `3/4ub4` (x >= 1/4, y > 2/3): gives psi(2/7,5/7) >= 3/4 = 21/28, consistent with the new upper bound 23/28 (so psi(2/7,5/7) is pinned to [3/4, 23/28]).
- `3/4psilb` (the paper's psi < 3/4 regions): none of its four bullets covers (2/7, 5/7) or (5/7, 2/7) — no conflict.
- `2+3/3+2` (psi(x,1/3) >= 2/3 for x > 1/2, psi(1/3,y) >= 2/3 for y > 1/2): with the trivial bound these give psi(1/3,2/3) = psi(2/3,1/3) = 2/3, matching the Lemma's q = 3 case (where the writeup's two bounds coincide and no asymmetry arises — the writeup's condition q >= 7 is exactly right, the gap (q-5)/(q(q+1)) vanishing at q = 5).
- `peacepsi` (paper Thm 10.2, the paper's own partial symmetry: psi(x,y) = x iff psi(y,x) = x): since psi(5/7,2/7) = 6/7 != 5/7, it forces psi(2/7,5/7) > 5/7 — consistent with [3/4, 23/28], and confirms the counterexample lives precisely where `peacepsi` has no grip (neither value equals max(x,y)).

- Literature search (Sept 2026): no follow-up resolving psi-symmetry was found; the EJC 2022 version still lists the question as open, matching the catalog's review. The Lemma's exact value psi((q-2)/q, 2/q) = 1 - 1/q does not appear in the paper (its best applicable bound there is 3/4), so the result is new, not ALREADY_KNOWN.

Sources: arXiv:1902.10878, EJC v29i2p47, Princeton copy.

## B.4 Computational check

Scripts in `verification/scripts/1902.10878__01/` (plain python3, exact `fractions` arithmetic, no external packages):

- `verify_constructions.py` — rebuilds both constructions independently from the writeup's prose and checks every degree condition against the paper's definitions (stability and no A–C edges hold by construction and are respected in the distance-2 computation). Output:
  - Section-3 construction, q = 7: |A| = 28, |B| = |C| = 7, biconstrained = YES for (2/7, 5/7), |N^2_A(c)| = 23 for **every** c in C, ratio exactly 23/28. q = 9: |A| = 45, all c reach 38 (38/45). q = 11: |A| = 66, all c reach 57 (19/22). All match the claimed 1 - 2(q-2)/(q(q+1)).
  - Cyclic construction, q = 3,5,7,9,11,13: biconstrained = YES for ((q-2)/q, 2/q), every c reaches exactly q-1 of q, ratio 1 - 1/q. For q = 7: 6/7 = 24/28.
  - ALL CHECKS PASSED.

- `verify_lemma_core.py` — machine check of the finite core of the Lemma's lower bound: (a) exact-fraction verification that two terms > 1/q exceed 2/q (forcing |I(b)| = 1); (b) exhaustive enumeration of all matchings of K_q for q = 3..13 confirming none covers all q vertices (4, 26, 232, 2620, 35696, 568504 matchings respectively; sanity check at q = 6 correctly finds a perfect matching); (c) exact re-derivation of every displayed identity (|A| = q(q+1)/2, q+1 = (2/q)|A|, gap = (q-5)/(q(q+1)), 1 - 10/56 = 23/28 < 24/28 = 6/7). ALL CHECKS PASSED.

No claimed property failed; every number in the writeup reproduced exactly.

## B.5 Caveats

- The Lemma's lower bound psi(5/7,2/7) >= 6/7 is a statement over all biconstrained graphs and cannot be brute-forced; it rests on the hand-checked rigidity argument (steps 2.1–2.6). I re-derived each forced equality myself and found no gap; the final obstruction is machine-verified. This is the one load-bearing part not reducible to a finite certificate.
- The writeup determines psi(5/7,2/7) = 6/7 exactly but only bounds psi(2/7,5/7) from above (23/28); combined with the paper's `3/4ub4` the true value lies in [21/28, 23/28]. The disproof needs only 23/28 < 24/28, which is airtight; the writeup's caveat says this correctly.
- The asymmetry genuinely requires q >= 7: at q = 3 and q = 5 the two bounds coincide (gap (q-5)/(q(q+1)) = 0 or negative), consistent with the paper's known equalities at (1/3, 2/3). No off-by-one issues found.
- Convention dependence checked: "distance exactly two" vs "common B-neighbour" coincide here because tripartitions forbid A–C edges and internal edges; the writeup's constructions satisfy the tripartition requirements (nonempty stable parts, partition of V).
- Divisibility edge cases in the Lemma (whether (2/q)|B|, (2/q)|C|, |A|/q are integers) are all handled by the forced-equality structure, as noted in step 2.1.
- Untouched by the writeup, and correctly flagged as such: phi-symmetry (paper's Thm 2.3) is unaffected, and the exact-case (xi) symmetry question remains open. The construction's three A-vertices complete to B mean it says nothing about the exact-degree regime.

### B.6 Referee summary

I set out to break this proof and could not. The claimed counterexample splits into a finite certificate and one infinite-family lemma. The finite certificate — a 42-vertex (2/7,5/7)-biconstrained triple with |A| = 28, |B| = |C| = 7 in which every C-vertex reaches exactly 23 of the 28 A-vertices — was rebuilt independently from the prose and verified exactly (as were the q = 9, 11 members of the family and the cyclic constructions), against the definitions taken verbatim from the arXiv v3 source of 1902.10878. The lemma (psi(5/7,2/7) = 6/7) I verified line by line: the disjointness/equality rigidity, the edge-count, and the packing inequalities are all correct, and the concluding parity obstruction was additionally machine-checked by exhaustive matching enumeration. The two values are consistent with every theorem the source paper proves (trivial bounds, the 3/4-threshold theorems, and notably `peacepsi`, the paper's own partial symmetry result, which the counterexample cleanly evades), the question is confirmed open in the literature as of September 2026, and the interpretation is exactly the authors' question. The symmetry psi(x,y) = psi(y,x) is disproved.

## References

[1] M. Chudnovsky, P. Hompe, A. Scott, P. Seymour and S. Spirkl, Concatenating bipartite graphs, *Electron. J. Combin.* 29(2) (2022), P2.47; arXiv:1902.10878.